

Week 8 Submission

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November 26, 2025

1 Introduction

There's an often told story of Hippasus, a mathematician during Pythagoras's time, that found out that the length of the diagonal of a square was not commensurate with the length of its side. He was apparently drowned for this discovery by the Pythagoreans. At some level, I can sympathize with the Pythagoreans, since this proof is deeply unsettling when I first read it. Like, rational numbers can be made as small and accurate as you want, what do you mean there's something it can't perfectly describe? And even after learning real analysis, the way $\sqrt{2}$ is actually described (whether it's a Dedekind cut, equivalence class of Cauchy sequences, or something else) leaves much to be desired. I much prefer the algebraic treatment of $\sqrt{2}$ instead, where just like i , we just claim it's not a part of the rationals and adjoin it, without making any claims about what it really is. So my proof won't say that $\sqrt{2}$ is irrational, but rather that the rational numbers do not have such a number.

2 The Proof

Theorem 2.1.

$$\forall q \in \mathbb{Q}, q^2 \neq 2.$$

Proof. Assume for contradiction that there does exist a number q in the rationals whose square is 2. We use the property that all rational numbers can be written in the most reduced form (proof of this property left to the reader) to write

$$q = \frac{a}{b},$$

where $a, b \in \mathbb{Z}$, $b \neq 0$, and a and b are coprime. Thus

$$q^2 = 2 = \frac{a^2}{b^2},$$

which implies

$$2b^2 = a^2.$$

We use the property that the product of even numbers is even and the product of odd numbers is odd (proof left to the reader) to conclude that a must be even since a^2 is even. Thus we can write $a = 2c$ for some $c \in \mathbb{Z}$. We then have

$$q^2 = 2 = \frac{(2c)^2}{b^2},$$

which implies

$$2b^2 = 4c^2.$$

Dividing both sides by 2, we have

$$b^2 = 2c^2,$$

which using the latter property again tells us b is even. But this contradicts the fact that a and b are coprime since they both share a factor of 2. $\square$